The reaction quotient Q at this stage,

$$Q = \frac{[HI]^2}{[H_2][I_2]} = \frac{1.092 \times 1.092}{0.154 \times 0.154} \approx 50$$

The Q value is equal to Kc value.

So, we can conclude that the system has reestablished the equilibrium at this stage and the addition of iodine results in the increase in the HI concentration.

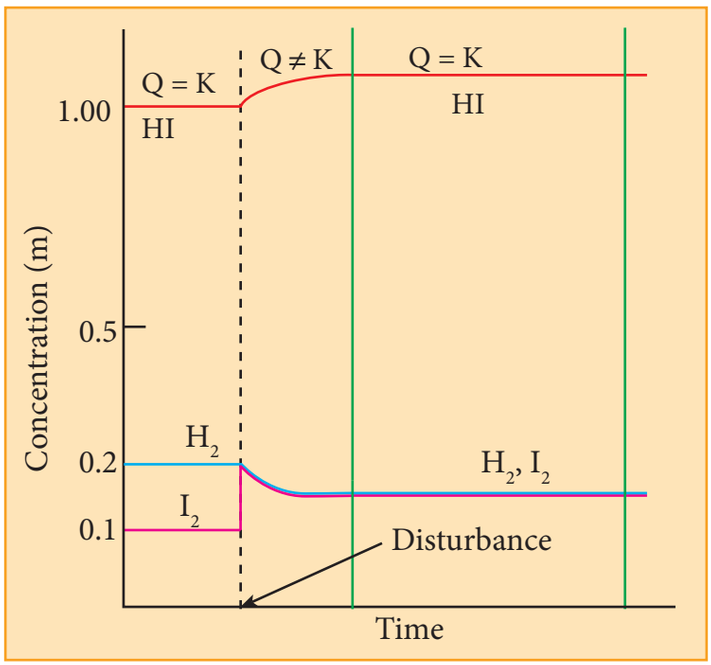


Figure 8.5 Effect of addition of iodine on formation of HI

DO YOU KNOW?

In the large scale of production of CaO from $CaCO_3$, constant removal of CO_2 from the kiln drives the reaction to completion. As in the case of manufacture of NH_3 by the Haber's process, NH_3 is liquefied and removed so that the reaction keeps moving in forward direction.

8.8.2 Effect of pressure

The change in pressure has significant effect only on equilibrium systems with gaseous components. When the pressure on the system is increased, the volume decreases proportionately and the system responds by shifting the equilibrium in a direction that has fewer moles of gaseous molecules.

Let us consider the synthesis of ammonia from nitrogen and hydrogen.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

11th Chemistry Chapter-8 Vol-2 EM.indd 17 09-01-2023 14:38:28